

Genetic Algorithm for Multi-Stop Delivery Route Optimization

A Practical Research Topic on Combinatorial Optimization

Topic 1 • Algorithms & Optimization¹

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Abstract

A driver or delivery vehicle must visit many locations while minimizing total travel distance, travel time, fuel cost, or late deliveries. This problem is closely related to the Traveling Salesman Problem (TSP) and the Vehicle Routing Problem (VRP). This document proposes a practical research project that investigates whether a genetic algorithm can efficiently find near-optimal multi-stop delivery routes under real-world constraints such as distance, delivery deadlines, and vehicle capacity. It outlines the background, the core research question and sub-questions, a five-step research plan, the expected outcome, and source references.

1. Basic Background

Multi-stop delivery routing is a classic optimization problem. A driver or delivery vehicle must visit many locations while minimizing total travel distance, travel time, fuel cost, or late deliveries. This problem is closely related to the **Traveling Salesman Problem (TSP)** and the **Vehicle Routing Problem (VRP)**.

A simple shortest-path algorithm such as Dijkstra can find the shortest path between two locations, but it does not directly solve the harder question: **what is the best order to visit many delivery stops?** As the number of stops increases, the number of possible routes grows extremely quickly. For example, 10 stops already create millions of possible orderings.

A **genetic algorithm** is useful because it does not need to check every possible route. Instead, it starts with many possible routes, evaluates them using a fitness function, keeps better routes, combines parts of good routes, adds small mutations, and gradually improves the solution.

2. Research Question

Can a genetic algorithm efficiently find near-optimal multi-stop delivery routes under real-world constraints such as distance, delivery deadlines, and vehicle capacity?

Possible sub-questions

1. How close is the genetic algorithm's route to the best-known or baseline route?
2. How does performance change as the number of delivery stops increases?
3. How do constraints such as time windows or vehicle capacity affect route quality?
4. Can a hybrid method using Dijkstra plus genetic algorithm improve performance?

3. Brief Research Plan

1. Create or collect delivery-location data

- Use a small city map, school campus map, or simulated delivery points.
- Each location has coordinates, demand size, and optional delivery deadline.

2. Build baseline algorithms

- Nearest-neighbor greedy algorithm.
- Random route generation.
- Optional: exact brute-force search for very small cases.

3. Implement the genetic algorithm

- Each chromosome represents one possible delivery order.
- Fitness function measures total distance, lateness penalty, and capacity penalty.
- Use selection, crossover, and mutation to evolve routes.

4. Optional hybrid design

- Use Dijkstra to compute shortest-path distances between all pairs of delivery locations.
- Use genetic algorithm to optimize the visiting order.

5. Evaluate performance

- Compare route distance, runtime, number of late deliveries, and scalability.
- Test with 10, 20, 50, and 100 stops.

4. Expected Outcome

The project should show that genetic algorithms are practical for large multi-stop delivery problems where exact search is too slow. The student can also build a visualization showing how the route improves over generations.

References

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<https://pmc.ncbi.nlm.nih.gov/articles/PMC5676484/>
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